

Anthony Ibarra | Electrical & Computer Engineer

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Summary

Electrical and Computer Engineering graduate student willing to learn and gain knowledge and techniques to grow as an engineer. Striving to try new things and embrace them to grow and become a part of something remarkable. Strong interest in Control, Robotics, power electronics, and Embedded Systems. I enjoy finding ways to improve my skill set either in new or old implementations.

EDUCATION

University of Illinois Chicago (UIC)

Jan 2023 - May 2025

Master of Science in Electrical and Computer Engineering

Relevant Coursework: Audio & Acoustic Signal Processing, Adaptive Digital Filters, Linear Systems Theory & Design, Advanced Computer Communication, Convex Optimization, Intro Neural Networks

University of Illinois Chicago (UIC)

Aug 2017 - May 2022

Bachelor of Science in Computer Engineering

Relevant Coursework: Artificial Intelligence I, Principles of Modern Control & Principles of Auto Control, Pattern Recognition I, Computer Comm Networks I, Embedded Systems

SKILLS

Computer programming: Python, C/C++/C# , Ubuntu, AI (artificial intelligence) & ML(machine learning)

Software Knowledge: VS Code, GitHub, MATLAB, Quartus, code-composer, Arduino IDE, VNC viewer(raspberry Pi), (Microsoft Word, PowerPoint, Excel, etc.)

Speech: B1 Proficient in spanish

Academic Projects

Self Driving Car, Mechatronics Embedded Design

Jan 2023 - May 2023

- Lead the team and manage all time constraint task, development and design task, make timely decisions for the team's success
- Develop a motor controller with a FET Driver implementation, either of (single fet, half bridge, or full H-bridge), and gets controlled via a PWM input signal from the microcontroller
- Develop a Boost Converter DC-DC for the power systems
- Develop and tune a PD & P controller for the steering(servo motor), and velocity controller(dc motor) respectively
- Design a circuit through Altium Designer and get it manufactured, create a perf board circuit as a backup board for pcb failure
- Implement Sensors and Encoders Implement a filter for the line Camera or velocity measured input

Wildfire Classification, Intro to Neural Networks

Jan 2023 - May 2023

- Using Python with Pytorch as the framework to develop a neural network model
- Create an image classification model, and adjust the model for current needs for dataset and computation constraints

Automated Watering System, Senior Design

August 2021 - May 2022

- Work in a four-student team to prototype a device that water specified plants by taking moisture levels, outdoor weather conditions, and plant information into account.
- Manage team to make sure all assignments are done on time and completed, and submit weekly assignments based on progress of project development.
- Program Arduino Nano iot 33 to decide whether to water or not water plants based on moisture levels
- Create APP (Kivy framework was used) to show information regarding the system, plants and weather
- Program a UDP client-server communication between Arduino and APP